

Program : Diploma in Printing Technology	
Course Code : 2101	Course Title: Basics of Printing Processes
Semester : 2	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To develop knowledge on different stages of printing process.
- To provide the concept of working principle of conventional printing methods.
- To impart the knowledge of basic units of a printing machine.
- To illustrate digital printing process.

Course Prerequisites:

Topic	Course code	Course name	Semester
Visible spectrum, Viscosity, Surface tension.		Applied physics I	1
Chemical affinity, Emulsion, Coating, pH, Conductivity, Etching.		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the stages of printing process.	10	Understanding
CO2	Interpret basic types of printing processes.	11	Understanding
CO3	Describe the workflow of Letterpress, Offset, Flexography and Gravure printing process.	13	Understanding
CO4	Describe the features of screen-printing and Digital printing process.	9	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the stages of printing process.		
M1.01	To define printing.	1	Remembering
M1.02	To explain the history of Printing.	3	Understanding
M1.03	To differentiate – Press, Pre press, Post press.	4	Understanding
M1.04	To identify various printed products.	2	Understanding
Content: Printing –definition. History of printing - from early stone print to latest innovations. Classification of print works - Prepress, Press, post press. Various printed products.			
CO2	Interpret basic types of printing processes.		
M2.01	To define the terms and Impact & Non-impact printing.	2	Understanding
M2.02	To explain the principle of Relief printing process.	2	Understanding
M2.03	To explain the principle of Planographic printing process.	2	Understanding
M2.04	To explain the principle of Intaglio printing process.	2	Understanding
M2.05	To explain the principle of Stencil printing process.	2	Understanding
M2.06	To define the digital printing process.	1	Understanding
	Series Test – I	1	

Content: Impact (Conventional) and non-impact printing – definition. Basic principles of - relief printing), Planography (lithography, offset), stencil printing and digital printing processes. (letter press, flexography), intaglio printing (gravure).

CO3	Describe the workflow of Letterpress, Offset, Flexography and Gravure printing process.		
M3.01	Basics of Letter press.	3	Understanding
M3.02	Units of Offset Printing.	3	Understanding
M3.03	Units of gravure printing machine.	3	Understanding
M3.04	Units of flexographic printing machine.	4	Understanding

Contents: Functional units of Conventional printing machines-Basics of Letter press (Introduction and workflow only). Units of Offset Printing- sheet fed and web offset Printing, plate cylinder, Blanket Cylinder, Impression cylinder, inking and dampening system. Units of gravure printing machine - gravure cylinder, impression roller, doctor blade. Units of flexographic printing machine - Fountain roller, Anilox roller, plate cylinder, impression cylinder.

CO4	Describe the features of screen-printing and Digital printing process.		
M4.01	Screen printing - Equipment and materials	3	Understanding
M4.02	Photographic stencil methods	3	Understanding
M4.03	Digital printing	3	Understanding
	Series Test – II	1	

Contents: Screen printing-definition. Equipment and materials: frames (wooden frames, metal frames, self-stretching frames). Mesh materials (silk, synthetic, stainless steel).Substrates- (paper, paper board, wood, textiles, plastics, metals, ceramics & glass). Digital printing basics.

Text / Reference:

T/R	Book Title/Author
T1	TulikaDas, Chemistry in Printing, 2nd Edition , BarnanaPrakashani, 2011.
R1	P. Kipphan, Handbook of Print Media , Springer, 2002.
R2	BishwanathChakravarty, A Hand Book for Printing and Packaging , Galgotia Publications Pvt. Ltd, 1997.

R3	N.R.Elred& T. Scarlet, What the Printer should know about Ink , GATF, 1995
R4	N.R.Elred& T. Scarlet, Chemistry for the Graphic Arts , GATF, 1992.
R5	R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition , Kluwer Academic publisher.

Online Resources:

Sl.No	Website Link
1	https://www.britannica.com/technology/offset-printing
2	https://www.britannica.com/technology/flexography
3	printwiki.org/Gravure
4	https://medium.muz.li/a-beginners-guide-to-screen-printing-by-a-complete-beginner-c5bd393d9238